

SegmentGDA: Training-Free Few-Shot Segmentation via Gaussian Discriminant Analysis on Vision Foundation Features

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Motivation

Segment a novel object from only a few annotated support examples.

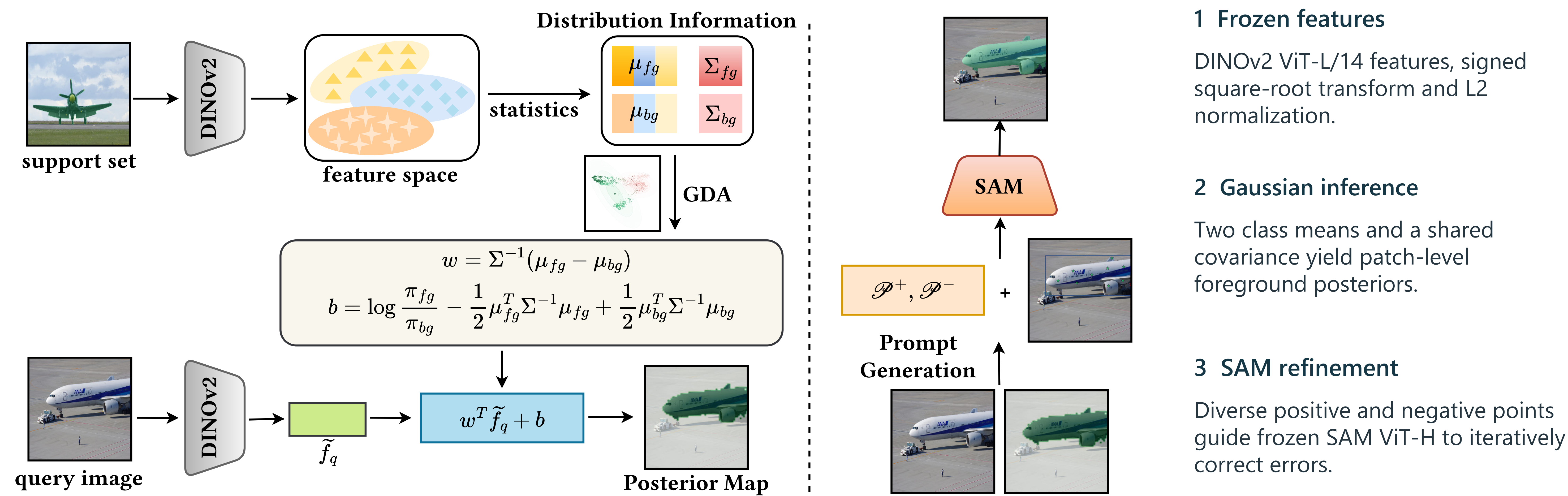
Local patch matching can miss object parts and select background clutter.

Core idea

Model foreground and background as distributions in frozen foundation features.

Bayesian posteriors guide SAM prompts. No task-specific training or fine-tuning.

Method overview



5-shot comparison with training-free methods

mIoU (%) Selected baselines from Tables 1-2 of the paper

Method	COCO-20 ⁱ	FSS-1000	LVIS-92 ⁱ	PASCAL-5 ⁱ
Matcher	60.7	89.6	40.0	74.0
UNICL-SAM	78.7	86.3	37.4	-
GF-SAM	66.8	88.9	44.2	82.6
SegmentGDA (ours)	62.5	89.8	41.8	82.2

Bold: best among the methods shown. -: not reported in our paper.

Higher than Matcher on all four benchmarks; competitive on FSS-1000 and PASCAL-5ⁱ.

Our 1-shot mIoU, in the same dataset order: 42.1 / 82.9 / 26.2 / 59.5.

Single-example performance remains a limitation; the main gains occur with multiple supports.

Qualitative comparisons



Rows: ground truth / Matcher / GF-SAM / SegmentGDA. Selected 5-shot episodes.

TAKEAWAY Global statistics improve multi-shot mask transfer.

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CODE